

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

(100)

Department of Chemical Engineering

University of Illinois Chicago

Midsemester Exam #2**Only one sheet of paper 8 1/2 x 11 is allowed****No phones or i-pads or computers are allowed****Duration: 70 minutes --- October 21, 2025****All Problems must be done****Problem 1: (50 points)**

Let's design a shell and tube heat exchanger to heat a stream with $m_c C_{p,c} = 5,000 \text{ BTU} / \text{h}^\circ\text{F}$ from 90°F to 210°F flowing in the tubes using as the heat source a stream with $m_H C_{p,H} = 10,000 \text{ BTU} / \text{h}^\circ\text{F}$ entering the shell at 260°F . The heat exchanger will contain tubes with a 0.5 in. outside diameter. In order to maintain a reasonable fluid velocity inside the tubes, $N_t = 30$ tubes per pass will be used. The overall heat transfer coefficient is $U_o = 150 \text{ BTU} / \text{h ft}^2 \text{ }^\circ\text{F}$. For a heat exchanger with one-tube pass and one-shell pass, determine the required tube length. **(40 points)**

Suppose that we need to have shorter tubes, what can we do? **(10 points)**

$$m C_{p,c} = 5000 \text{ BTU/h}^\circ\text{F}$$

$$Q = m C_{p,c} (T_{c,out} - T_{c,in}) = 5000 (210 - 90) = 600000$$

Energy Balance

$$Q = m C_{p,H} (T_{H,in} - T_{H,out}) \Rightarrow 600000 = 10000 (260 - T_{H,out})$$

$$T_{H,out} = 200^\circ\text{F}$$

$$\Delta T_1 = T_{h,in} - T_{c,out} = 260 - 210 = 50^\circ F$$

$$\Delta T_2 = 200 - 80 = 120^\circ F$$

$$\Delta T_{em} = \frac{\Delta T_2 - \Delta T_1}{\ln(\Delta T_2 / \Delta T_1)} = \frac{60}{\ln(120/50)} = 76.103^\circ F$$

$$A = \frac{Q}{U_o \Delta T_{em}} = \frac{500000}{150 \times 76.10} = 52.562 \text{ ft}^2$$

Out side area N_t , length L

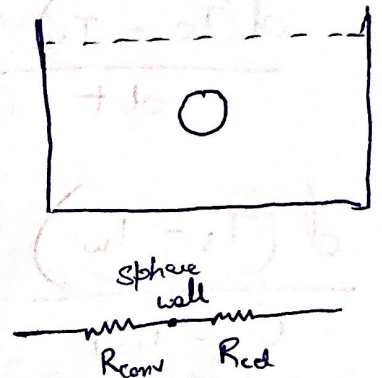
$$A = N_t \pi d_o L \Rightarrow L = \frac{A}{N_t \pi d_o} = \frac{52.562}{30 \pi (0.04)} = 13.4 \text{ ft}$$

Required tube length = 13.4 ft

L depends on A directly $\rightarrow$ decrease the Area
 N inversely $\rightarrow$ increase the count
 increase the U_o as A depends on it 50

Let's consider a case where 1 kg of aluminum spheres with radius $R = 10$ mm initially at 90°C are immersed in 1 L of water initially at 20°C in a tank. The tank is insulated and heat losses are negligible. The water is assumed to be perfectly mixed. Determine the time needed to cool the aluminum spheres to 40°C assuming that the external heat transfer resistance at the sphere-fluid interface is dominant and that $h = 500\text{ W / m}^2\text{ K}$. The density of aluminum is $\rho = 2,700\text{ kg / m}^3$ and its heat capacity is $C_P = 900\text{ J / kg K}$. Assume that the water temperature remains constant. **(30 points)**

→ General PDF =
(given in the question convective resistance $\gg$ conductive)
 $\therefore B_0 < 0.1$



$$m C_p \frac{dT_s}{dt} = -h A_s (T_s - T_\infty) \rightarrow \text{② balance on sphere}$$

$$\frac{dT_s}{dt} = \frac{-hA_s}{m_s C_p} (T_s - T_\infty)$$

Separation of variables

$$\int \frac{dT_g}{(T_g - T_{\infty})} = \int_0^t - \frac{h A_s}{m_s C_p} dt$$

$$\ln \frac{T_{sf} - T_{\infty}}{T_{si} - T_{\infty}} = \frac{-h A_s}{m_s C_p} t$$

$$\text{or } t = - \ln \left(\frac{T_{s,f} - T_{\infty}}{T_{s,i} - T_{\infty}} \right) \frac{m_s C_p}{h A_s} \quad 30$$

$$V_{\text{total}} = \frac{m}{\rho} = \frac{1}{2700} = 3.7 \times 10^{-4} \text{ m}^3$$

Volume for one sphere

$$= \frac{4}{3} \pi R^3 = 4.19 \times 10^{-6} \text{ m}^3$$

$$\therefore N = \frac{V_{\text{total}}}{V} = 88.3$$

$$\therefore \text{Total Area} = N(4\pi R^2)$$
$$= 88.3 \times 4\pi (0.01)^2$$
$$= 0.111 \text{ m}^2$$

$$t = \frac{(1) (999)}{999 (0.111)} \times \ln \left(\frac{30-20}{40-20} \right) = 16.2 \times 1.25^{-3} \\ = \underline{\underline{20.3 \text{ sec}}}$$

$$(b) \quad m_s c_p (T_s - T_{s,i}) + m_w c_{pw} (T_w - T_{w,i}) = 0 \quad \checkmark$$

Lumped mass DE

As now water's temp is changing

$$m_s c_p \frac{dT_s}{dt} = -hA_s (T_s - T_w)$$

~~now~~ substitute to eliminate $(T_s - T_w)$

$$\frac{d(T_s - T_w)}{dt} = -hA \left(\frac{1}{m_s c_p} + \frac{1}{m_w c_{pw}} \right) (T_s - T_w)$$

$$\frac{d(T_s - T_w)}{(T_s - T_w)} = -hA \left(\frac{1}{m_s c_p} + \frac{1}{m_w c_{pw}} \right) dt$$

~~or~~

~~(T_s - T_w)~~